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Original Article

Evolutionary associations of bill morphology, migration, and feeding ecology in Tyrant flycatchers (Tyrannidae)

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Abstract

Migratory birds experience changes throughout their annual cycle in food availability and diet, which may influence the evolution of feeding morphology. We investigated the evolution of bill morphology in relation to migratory phenotype, diet, and foraging behavior across 386 species of Tyrant flycatchers (Tyrannidae), and among migratory and nonmigratory populations of the Fork-tailed flycatcher (*Tyrannus savana*). At the interspecific level, bill morphospace extensively overlapped between migrants and nonmigrants, with small differences in bill shape but not size. Migrants had deeper and narrower bills at inter- and intraspecific levels. However, this shape difference was not more pronounced between sister species differing in migratory phenotype, suggesting that the changes in migratory phenotypes during speciation were not associated with bill divergence. At the intraspecific level, migratory Fork-tailed flycatchers also had smaller bills varying according to age but not sex. Across Tyrannidae, species with an invertivore–herbivore diet that do not forage on the ground and understory exhibited wider and shorter bills. Overall, our results suggest a weak but significant association between bill morphology and migratory phenotypes, with additional variation in diet, foraging strata, and age. However, bill morphology appears to have evolved independently of migratory behavior, underscoring a decoupling of migration from foraging-related morphological constraints.

Keywords: correlated evolution, diet, feeding morphology, foraging, morphological evolution, migratory phenotype

Introduction

The evolution of behavior can drive correlated evolution of morphological traits. For instance, migratory behavior is an evolutionary adaptation to seasonal environments enabling organisms to track resource availability across space and time, which often requires modifications in morphology (Shaw, 2016). In birds, migration can involve journeys of up to thousands of kilometers to areas where resources are available during the nonbreeding season, with return flights when resource conditions become favorable again for breeding (Berthold, 2001; Newton, 2007; Winger et al., 2018). A well-known pattern is that migratory birds have longer and more pointed wings in comparison with their nonmigratory relatives, which likely enhances efficient sustained flight (Lockwood et al., 1998; Sheard et al., 2020). Moreover, before departing, many migratory birds increase their foraging rate to accumulate fat as fuel for long-distance movements, enabling them to cross ecological barriers such as the Caribbean Sea or the Saharan Desert (Alerstam et al., 2003; Bayly et al., 2018; Cano et al., 2020; Fransson et al., 2008; Newton, 2007). Also, many species switch from specialized to more generalist diets on their wintering grounds. For example, Eastern kingbirds (*Tyrannus tyrannus*) consume both fruits and insects in South America, whereas they are mainly insectivorous in the breeding grounds in North America (Kricher, 2017; Morton, 1980). Likewise, some migratory birds shift their trophic niche because they need to molt their plumage, and this requires large amounts of energy (MacPherson et al., 2021). In sum, the dietary changes experienced by migratory birds, combined with the physiological demands of sustained flight and molt, which often require increased foraging rates, may also influence the evolution of relevant traits such as feeding morphology (e.g., MacPherson et al., 2022).

Avian bills are multifunctional structures that evolve under different selective pressures including food handling, vocal signaling, thermoregulation, and social interactions (e.g., Crandell et al., 2019; Kelly et al., 2024; Podos, 2001; Rico–Guevara & Araya–Salas 2015; Tattersall et al., 2009). Consequently, a change in migratory strategy may be buffered or overridden by these ecological constraints, making it difficult to detect a clear association of bill morphology with migration. Thus, understanding the drivers of bill morphological variation requires considering potentially interacting effects of multiple selective pressures (Navalón et al., 2019), including diet type, food selection and manipulation, trophic niche breadth, and foraging behavior (Brodt &

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Cáceres, 2025; Gosler, 1987; Kurtz et al., 2024; Pigot et al., 2020; Sayol et al., 2025; Xu et al., 2023). Moreover, factors such as age and sex can also contribute to variation in bill morphology (Pyle, 2022a, 2022b). Ontogenetic changes in bill size and shape occur as individuals mature, potentially reflecting shifts in diet, functional abilities, or developmental constraints (Parker, 1994; Paton & Collins, 1989; Tattersall et al., 2009). Additionally, sexual dimorphism in bill morphology is often linked to differences in foraging behavior, territoriality, or reproductive roles, and it is also observed in multiple avian species (Paton & Collins, 1989; Székely et al., 2000).

One approach to studying how differences in migratory behavior can influence bill morphology is the analysis of "morphospace", a multivariate representation of morphological variation, in which axes typically correspond to linear measurements or shape components derived from geometric morphometrics (Budd, 2021; Pigot et al., 2020). In this space, bills of avian generalists, with greater intraspecific morphological variation, often tend to cluster near the center, likely reflecting functionally efficient trait combinations for exploiting a wide variety of resources, whereas specialists, with smaller intraspecific morphological variation, tend to occupy peripheral regions representing more extreme trait combinations that allow them to exploit a smaller variety of resources (Pigot et al., 2020; Van Valen, 1965). We hypothesize that migratory birds will tend to occupy central positions in bill-morphospace and show somewhat higher within-species variation because long-distance movement requires dietary flexibility across diverse stop-over habitats, favoring a more generalist bill shape for multiple food types encountered in route. Alternatively, if migration is tightly linked to specialized diets, their bills could occupy the periphery with less variation.

The highly speciose Tyrant flycatcher family (Tyrannidae) is primarily composed of insectivorous species that capture prey through multiple types of foraging behaviors, including aerial hawking, sally-gleaning, perch-to-ground sallying, perch-gleaning, and ground foraging (Fitzpatrick, 1981; Sherry, 1984; Winkler et al., 2020), some of which are correlated with bill size (Fitzpatrick, 1985). Flycatchers have a broad geographic distribution in the Americas and exhibit a variety of migration phenotypes (Winkler et al., 2020). Across their evolutionary history, migration has been frequently lost, seemingly promoting the origin of some nonmigratory species from migratory ancestors (Gómez-Bahamón et al., 2020a). For instance, a group of Fork-tailed flycatchers (*Tyrannus savana*) lost migration in the wintering grounds, a process resulting in various life-history changes leading to reproductive isolation from the migratory ancestral population and significant differences in flight morphology (Gómez-Bahamón et al., 2020a) and communication signals (Gómez-Bahamón et al., 2020b). Repeated shifts in migratory behavior across the family and those within Fork-tailed flycatchers offer an opportunity to study the correlated evolution of behavior and morphological traits at the inter- and intraspecific level.

We aim to understand if there is a correlated evolution of bill morphology with shifts in migratory phenotype while also accounting for dietary and foraging strategies. We tested whether the ecological demands of migration favor more generalized bill shapes in migratory species than in their nonmigratory close relatives. Because birds must build large fat reserves in a short predeparture window and encounter fluctuating food availability throughout the year, a bill that can handle a wide range of dietary items could enable faster energy intake and fat accumulation. Consequently, we predict that migrants will tend to have more centrally positioned (i.e., generalist) bill morphologies in morphospace compared with their nonmigratory counterparts. To further evaluate the association between migratory behavior and bill size and shape, we tested whether morphological divergence is higher between sister species pairs that differed in migratory phenotype compared to those in which there is no difference in migratory phenotype. Finally, we also examined whether intraspecific variation of bill morphology in migrant and nonmigrant Fork-tailed flycatchers evolves in accordance with that observed at higher taxonomic levels (with respect to variation and position in the morphospace), while evaluating associations between bill morphology and migratory phenotype, accounting for age and sex.

Materials and methods

Interspecific level data collection (Aves, Tyrannidae)

We conducted comparative phylogenetic analyses using a trimmed version of a comprehensive phylogeny of the suboscine passerines (Harvey et al., 2020). We obtained mean bill measurements (bill depth, culmen length from nares, and bill width) for 423 species of Tyrant flycatchers from the Avonet dataset (Tobias et al., 2022). We also obtained migratory phenotype, employing the variable as dichotomous: 0 = nonmigratory (i.e., sedentary) vs. 1 = migratory (i.e., "partial-migratory" [minority of population migrates long distances, or most of population undergoes short-distance migration, nomadic movements, distinct altitudinal migration, etc.] + "migratory" [majority of population undertakes long-distance migration] birds classified in the same Avonet dataset) following Tobias et al. (2022). Additionally, we used the dietary and foraging strategies of birds described in Wilman et al. (2014), classifying species in three categories of diet: (1) Omnivores, (2) Invertivores, and (3) Invertivores–herbivores; and five categories of foraging strata: (1) canopy, (2) ground, (3) ground–understory, (4) understory, and (5) upper branches. After removing species with missing data for morphological, diet, or behavioral variables, our data set comprised 386 species.

Intraspecific level data collection (Fork-tailed flycatcher)

We captured Fork-tailed flycatchers in two study sites in Argentina (Reserva Natural El Destino, 36°79′51″ S, 64°28′27″ W, Buenos Aires province; and Reserva provincial Parque Luro, 35°08′21″ S, 57°23′19″ W, La Pampa Province), two in Brazil (Estação Ecológica de Itirapina, 22°18′0″ S, 47°53′59″ W, São Paulo State; and Parque da Alvorada, 15°48′0″ S, 47°47′59″ W), and one in Colombia (Reserva Natural Tomogrande, 4°50′12″ N, 70°12′56″ W, Vichada Department) using methods described in Jahn et al. (2016). In total, we captured 218 individuals, corresponding to 163 migratory *T. s. savana* and 55 nonmigratory *T. s. monachus*, in 2013, 2014, 2015, and 2019.

We identified individuals by plumage in the areas where they co-occur in Colombia, given that *T. s. monachus* has a lighter gray back and a fuller white collar compared to *T.*

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s. savana, and adult males of *T. s. monachus* have two narrowly notched outer primaries, while *T. s. savana* has three (Pyle, 2022b). We banded flycatchers with a unique metal-numbered band and measured bill depth, culmen length, and bill width using an SPI Polymid Dial Caliper 150 mm (±0.1 mm). We also recorded age (categorized as juvenile, immature, or adult) and sex according to molt and plumage stages (Jahn & Tuero, 2020; Jahn et al., 2016; Pyle, 2022b). We were not able to sex juvenile birds, so sample size by age and sex differed.

Data analysis

In comparisons within Fork-tailed flycatchers, we ran Principal Component Analysis on all bill size measurements. To visualize the morphospace across the Tyrant flycatcher family (386 species), we ran a phylogenetic PCA using the *phytools* package (Revell, 2024). We used PC1 (90.4%) at the interspecific level and PC1 (67.28%) at the intraspecific level as proxies for size in subsequent analyses (Table S1). We also estimated bill shape by calculating pairwise ratios among the three size measurements. Specifically, bill depth divided by bill width (depth/width), bill depth divided by culmen length (depth/length), and bill width divided by culmen length (width/length). Because morphological differences must be interpreted relative to body size through allometric relationships, tarsus length was included as a covariate in all models to control for body-size effects and assess variation in bill morphology independent of differences in overall body size.

To test whether the mode of evolution of migratory behavior and bill morphology is similar in Tyrant flycatchers, we quantified phylogenetic signal using Pagel's λ by employing the function *phylosig* in the *phytools* package (Pagel, 1999; Revell, 2024). Then, to analyze the intraspecific level association between bill morphology (size and shape), migratory phenotype, diet, and foraging strata, we ran linear models (LMs) incorporating phylogenetic relationships, using the *Phylolm* function in the *phylotools* package (Housworth et al., 2004; Revell, 2024). Specifically, we evaluated all possible combinations of the four independent variables: migratory phenotype, diet, foraging stratum, and tarsus length ($2^4 = 16$ candidate models) for each of the four bill morphology dependent variables (PC1 of bill size and the three bill shape ratios), resulting in 64 candidate models in total (Table S2).

To determine whether switches in migratory phenotype result in differentiation in bill morphology, we compared morphological divergence in the following sister-pair groups: (1) migratory and migratory sister clades, (2) nonmigratory and migratory sister clades, and (3) nonmigratory and nonmigratory sister clades. To obtain these sister groupings, we used the *getDescendants* function implemented in the *phytools* package (Revell, 2024). We calculated a morphological divergence index between sister pairs for bill depth/width because this shape ratio showed a significant association with migratory phenotype (see Results). To obtain the divergence index, we computed the ratio of the smaller to the larger mean value for each trait for the species in each pair comparison, obtaining a value between 1 and 0. Values closer to 0 correspond to higher morphological differentiation, and values closer to 1 to more similar morphology. For visualization purposes, we subtracted the obtained divergence index from 1 so that higher values corresponded to higher differentiation and lower values to lower differentiation. Then, to analyze the association between divergence events between clades that differ in migratory phenotype (i.e., between the three sister species pair groups) with divergence time and the effect on bill evolution, we ran four LMs using *the lm* function. We employed the divergence index (bill depth/width) of sister species pairs as the dependent variable, and migratory phenotype and divergence time as independent variables (Table 2). We obtained the divergence time (Ma) of sister species pairs using the function *branching.times* in the *ape* package (Paradis & Schliep, 2019).

We ran 64 models to analyze intraspecific associations between bill size and shape of Fork-tailed flycatchers according to their migratory phenotype and tarsus length while considering age and sex (Table S3). We ran LMs and mixed linear models (MLMs) with age and sex as random variables separately, using the function *lm* (Everitt & Hothorn, 2009) or *lmer* of the *lme4* package (Bates et al., 2015; Zuur et al., 2009), respectively. We were not able to analyze age and sex in the same models because juveniles were not sexed. In these LMs and MLMs, we evaluated all possible combinations using PC1 (67.3%) scores of bill size and bill shape ratios as dependent variables, and migratory phenotype (migratory *T. s. savana* vs. nonmigratory *T. s. monachus*) and tarsus length as independent variables (Table S3).

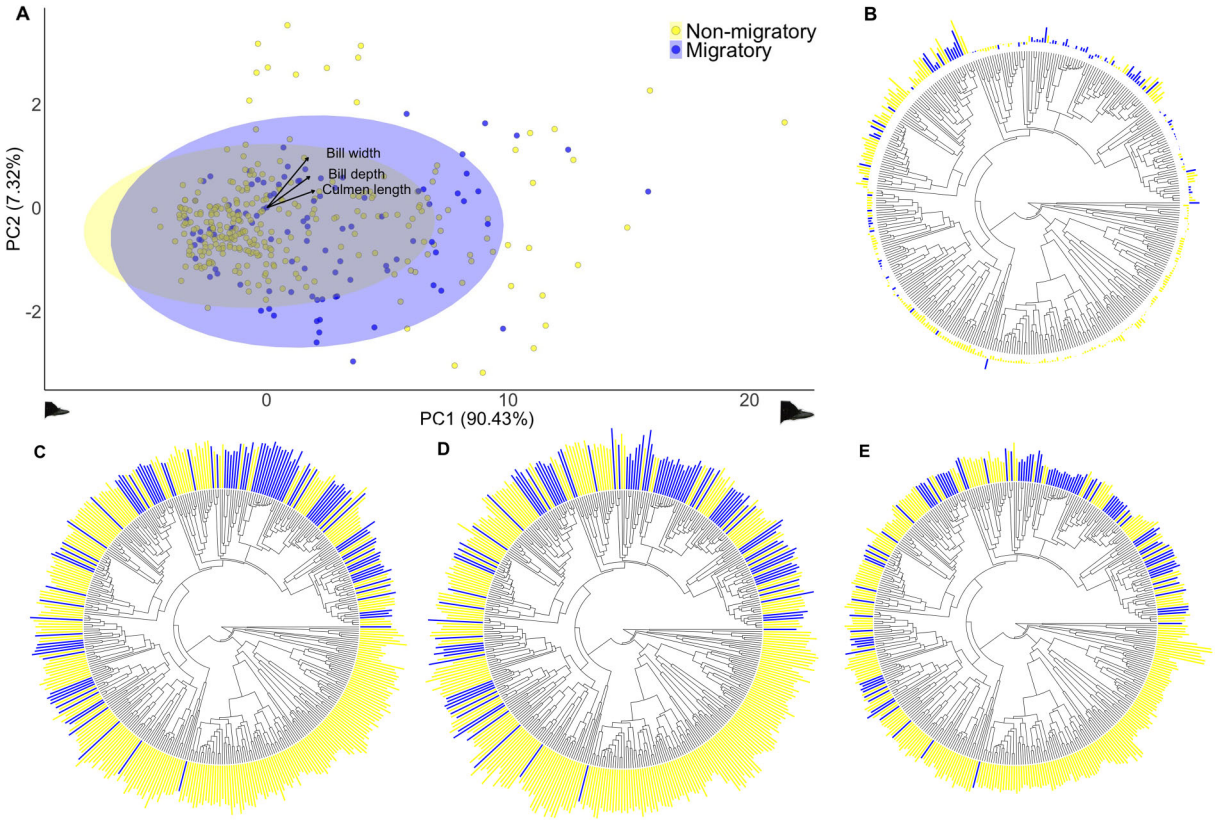
To normalize PC1 scores of bill sizes and measures of bill shapes, we applied a Yeo–Johnson transformation, which can be applied to variables that include zero or negative values (Yeo & Johnson, 2000). For model selection, we considered a model to be substantially better if its Akaike information criterion (AIC) scores were at least two units lower than its respective null model (Burnham & Anderson, 2004). To determine the significance of independent variables in the best-fitting models, we examined the 95% confidence intervals of their coefficients, considering variables statistically significant at the 0.05 level if their confidence intervals did not include zero (Burnham et al., 2011). All analyses were done using R version 4.2.2 (R Core Team, 2022), and all graphics were obtained using the *ggplot2* package (Wickham, 2016).

Results

Interspecific level: phylogenetic signal, morphospace, and best-fitting models

Our analysis shows that bill morphology is strongly phylogenetically structured while migratory status is labile, indicating that the two traits evolve independently. We found that phylogenetic signals for each bill size (depth $\lambda = 0.87$, length $\lambda = 0.91$, width $\lambda = 0.94$, PC1 (90.4%) $\lambda = 0.91$) and shape (depth/width $\lambda = 0.88$, depth/length $\lambda = 0.86$, width/length $\lambda = 0.97$, PC2 (7.3%) $\lambda = 0.93$) variables were high. Meanwhile, the presence of migratory phenotype had a low phylogenetic signal ($\lambda = 0.49$). PC1 and PC2 scores of bill morphology in migrants occupied similar positions in morphospace and extensively overlapped with those of nonmigrants (Figures 1A and S1). The best-fitting model for PC1 was the model that had "Tarsus" as an independent variable, whereas for depth/width was the model that had "Migration + Tarsus" as independent variables. The best-fitting models for depth/length were the models that had "Migration + Diet + Tarsus" and "Diet + Tarsus" as independent

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	B) PC1	C) Bill depth/width*	D) Bill depth/length	E) Bill width/length
Non-migratory	-4.65 — 21.4	0.43 —1.23	0.29 — 0.70	0.33 — 1.25
Migratory	-3.34 — 15.8	0.57 — 1.13	0.29 — 0.62	0.32 — 0.74

Figure 1. Migratory and nonmigratory Tyrant flycatchers (Tyrannidae) extensively overlapped in bill morphospace. Also, migrants showed deeper and narrower bills than nonmigrants. (A and B) Phylogenetic Principal Component Analysis of bill sizes and (C, D, and E) tree plots of bill shapes of Tyrant flycatcher species according to migratory phenotype. The asterisk indicates a significant association when we ran Phylogenetic Linear Models (Tables 1 and S2).

variables. Finally, the best-fitting models for width/length were the models that had "Diet + Tarsus", "Migration + Diet + Tarsus", "Diet + Foraging strata + Tarsus", and "Diet + Foraging strata + Migration + Tarsus" as independent variables (Table 1).

Interspecific level: shifts in migratory phenotype

Migratory phenotype was not associated with PC1, which corresponds to bill size (Figures 1A and 1B; Table 1). However, migrants showed significantly deeper and narrower bills (Figure 1C; Table 1) than nonmigrants, while migratory phenotype was not associated with bill depth/length (Figure 1D; Table 1) nor width/length (Figure 1E; Table 1).

Among the 121 sister-species pairs, 102 (84 %) shared the same migratory status, 83 were both nonmigratory, and 19 were both migratory, while the remaining 19 pairs comprised one migratory and one nonmigratory species (Figure S2). The depth/width differentiation index spanned 0.005–0.228 in migratory pairs, 0.000–0.238 in nonmigratory pairs, and 0.009–0.138 in mixed-status pairs (Figure 2A), with the greatest bill divergence occurring in a nonmigratory sister pair (Figure 2). Thus, species pairs that differ in migratory phenotype and those that are behaviorally the same did not differ in bill depth/width (Figure 2A), with none of the tested models being significant (Table 2). We also found that species pairs that have diverged for longer did not differ more in bill depth/width than those that recently speciated, and when we combined behavioral phenotype with divergence time, the relationship remained nonsignificant (Figure 2B; Table 2).

Interspecific level: dietary and foraging strategies

PC1 (90.4%) scores of bill size, bill depth/width, and bill depth/length were not associated with any diet or foraging category (Figures 3 and 4; Table 1). Bill width/length was positively associated with Invertivores–herbivores diet category and was negatively associated with ground and understory foraging category (Figures 3 and 4; Table 1).

Intraspecific level: morphospace, migratory phenotype, age, and sex

PC1 (67.3 %) and PC2 (24.2 %) scores for the migratory Fork-tailed flycatchers overlapped almost completely with those of nonmigratory (Figures 5A and S3A). When we analyzed PC1 (67.3%) scores of bill size and bill shape ratios

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Table 1. Best-fitting models (lower AIC and ΔAIC ≥ 2) using Phylogenetic Linear Models analyzing score 1 from Principal Component Analysis (PC1 = 90.4%) of bill size and measures of bill shape of Tyrant flycatcher (Tyrannidae) species according to their migratory phenotype, diet, foraging strata, and tarsus length. We used the Yeo–Johnson transformation to normalize the dependent variables, as appropriate. Bold text indicates: ΔAIC ≥ 2 for AIC values, 95% confidence intervals (Lower–Upper CI) for CIs that do not contain zero, and $p < .05$ for p values. Migratory phenotype categories: migratory and nonmigratory. Diet categories: omnivores, invertivores, and invertivores–herbivores. Foraging strata categories: canopy, ground, ground–understory, understory, and upper branches (see Table S2 for all results).

	Transf.	AIC	Lower–Upper 95% CI	*t*	*p*
PC1	Yes				
Null		1,366	−0.80–2.70	1.07	.29
Tarsus		**1,264**	**−0.33 to −0.23**	−10.96	**<.001**
Depth/width	No				
Null		−803	**0.66–0.85**	15.8	**<.001**
Migration + Tarsus		**−845**			
Migration			**0.01–0.06**	3.42	**<.001**
Tarsus			**0.007–0.01**	6.34	**<.001**
Depth/length	Yes				
Null		−2,308	**0.22–0.24**	35.6	**<.001**
Diet + Tarsus		**−2,325**			
Invertivores			−0.009–0.002	−1.14	.25
Invertivores–herbivores			−0.002–0.01	1.32	.19
Tarsus			**−0.001 to −0.001**	−3.24	**.001**
Migration + Diet + Tarsus		**−2,325**			
Migration			−0.001–0.005	1.24	.22
Invertivores			−0.008–0.003	−0.95	.34
Invertivores–herbivores			−0.002–0.011	1.42	.16
Tarsus			**-0.001 to –0.001**	−3.09	**.002**
Width/length	Yes				
Null		−2,327	**0.24–0.27**	33.4	**<.001**
Diet + Tarsus		**−2,380**			
Invertivores			−0.003–0.007	0.82	.41
Invertivores–herbivores			**0.001–0.012**	2.21	**.03**
Tarsus			**-0.002 to –0.001**	−7.26	**<.001**
Migration + Diet + Tarsus		**−2,380**			
Migration			−0.005–0.001	−1.36	.17
Invertivores			−0.004–0.007	0.59	.55
Invertivores–herbivores			**0.001–0.011**	2.07	**.04**
Tarsus			**−0.002 to −0.001**	−7.39	**<.001**
Diet + Foraging + Tarsus		**−2,379**			
Invertivores			−0.003–0.007	0.73	.46
Invertivores–herbivores			**0.001–0.012**	2.17	**.03**
Ground			−0.014–0.001	−1.84	.07
Ground–understory			**−0.013 to −0.001**	−2.04	**.04**
Understory			−0.01–0.002	−1.25	.21
Upper branches			−0.007–0.003	−0.74	.46
Tarsus			**−0.002 to −0.001**	−6.48	**<.001**
Migration + Diet + Foraging + Tarsus		**−2,379**			
Migration			−0.004–0.001	−1.15	.25
Invertivores			−0.004–0.007	0.56	.58
Invertivores–herbivores			**0.001–0.011**	2.05	**.04**
Ground			−0.014–0.001	−1.83	.07
Ground–understory			**−0.013 to −0.001**	−1.97	**.049**
Understory			−0.01–0.002	−1.27	.20
Upper branches			−0.007–0.003	−0.76	.45
Tarsus			**−0.002 to −0.001**	−6.58	**<.001**

according to migration types and tarsus length of the Fork-tailed flycatcher, and we also compared models considering age and sex as random factors, we found that only the PC1 model had an AIC < 2 with age as a random factor, while the best models of shape ratios were without random factors (Tables 3 and S3). Thus, migratory *T. s. savana* showed smaller PC1 scores of bill size considering age (Figures 5A and B; Table 3) and deeper and narrower bills (Figures 5C and S3B; Table 3) than nonmigratory *T. s. monachus*. Migratory phenotype was not associated with bill depth/length

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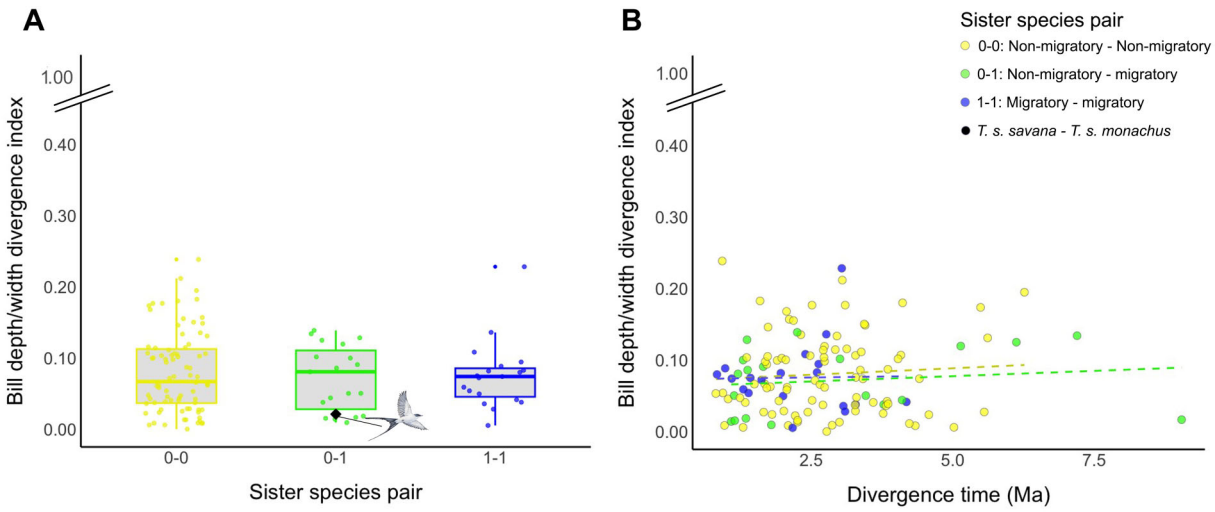


Figure 2. Bill shape divergence among Tyrant flycatchers (Tyrannidae) was not associated with migratory phenotype or divergence time. Divergence index of significant bill shape difference between Tyrant flycatcher sister species pairs according to (A) migratory phenotype and (B) divergence time. There were no significant associations in Linear Models (Table 2). 0–0: nonmigratory—nonmigratory, 0–1: nonmigratory—migratory, vs. 1–1: migratory—migratory phenotypes. Divergence indices of migratory *T. s. savana* and nonmigratory *T. s. monachus* are indicated using an illustration of a Fork-tailed flycatcher (*T. savana*) by Andreza da Silva, used with permission.

Table 2. Linear models analyzing divergence index of significant bill shape differences between sister species pairs of Tyrant flycatchers (Tyrannidae) according to migratory phenotype and divergence time. We used the Yeo–Johnson transformation to normalize the dependent variables, as appropriate. Bold text indicates: ΔAIC ≥ 2 for AIC values, 95% confidence intervals (Lower–Upper CI) for CIs that do not contain zero, and $p < .05$ for p values. Migration phenotypes are 0–0: nonmigratory—nonmigratory, 0–1: nonmigratory—migratory, vs. 1–1: migratory—migratory.

	Transf.	AIC	Lower–Upper 95% CI	*t*	*p*
Depth/width index	Yes				
Null		1,538	**367–411**	34.80	**<.001**
Migration		**1,542**			
Migration 0–1			−45.3–77.8	0.52	.60
Migration 1–1			−59.7–66.0	0.10	.92
Time		**1,540**	−21.3–10.9	−0.64	.53
Migration + Time		**1,543**			
Migration 0–1			−44.3–79.3	0.56	.58
Migration 1–1			−63.7–63.7	0.00	1.00
Time			−22.1–10.9	−0.67	.50

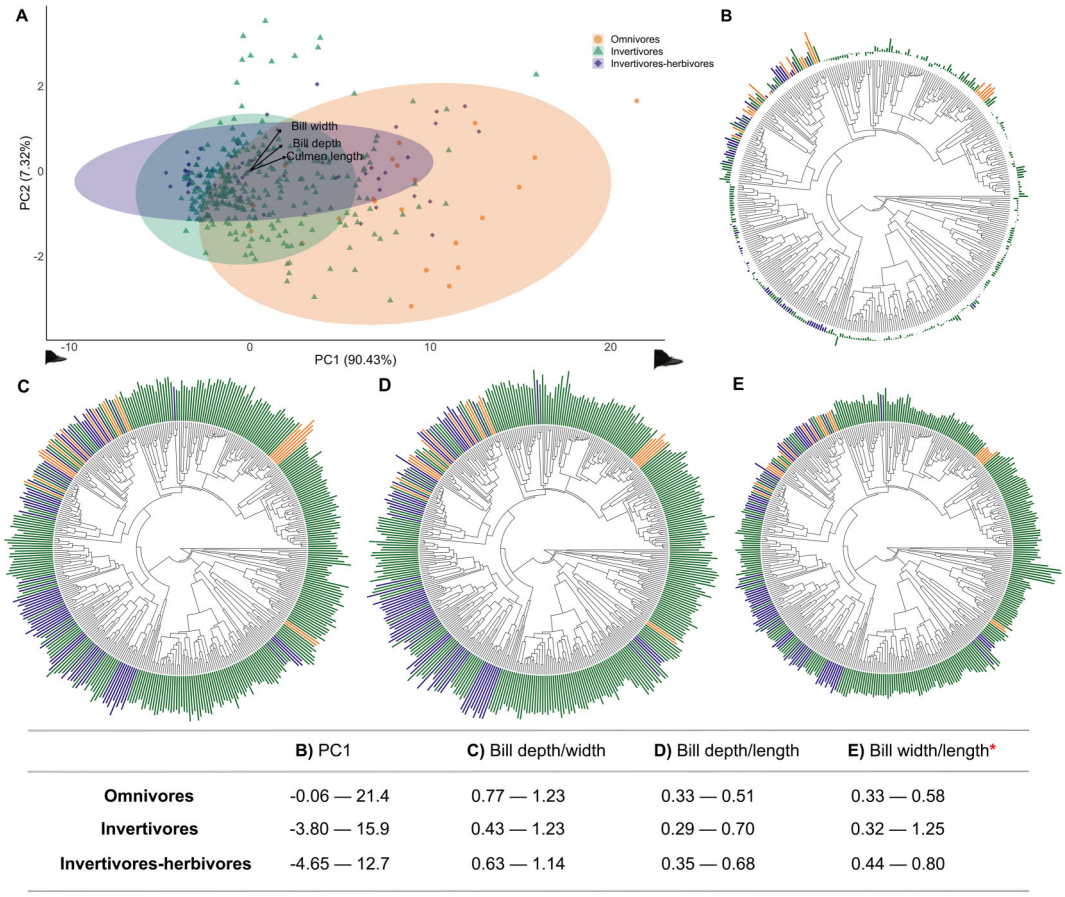
(Figure 5D) and width/length (Figure 5E) in the Fork-tailed flycatcher (Figure S3B; Tables 3 and S3).

Discussion

We found a small but significant difference in bill shape between migratory and nonmigratory Tyrant flycatchers. Migrants exhibited a deeper and narrower bill shape compared to nonmigrants at both inter- and intraspecific levels, but there was no evidence of occupying a more generalist morphology in morphospace than nonmigratory Tyrant flycatchers, despite the expectation that migrants would evolve more generalist bills to exploit the diverse food resources encountered along their migratory routes. One possible explanation is that this pattern could be reflecting aerodynamic pressures and not dietary associations. It is likely that a comparatively narrower bill slightly improves windbreakage during sustained flight, thereby reducing aerodynamic drag relative to broader-bill morphologies. However, we also found that speciation events that involved divergence into a migratory and a nonmigratory species did not result in higher divergence in bill shape compared to speciation events that involved two migratory or two nonmigratory lineages. Speciation events in Tyrant flycatchers that involve divergence in migratory phenotype tend to occur through loss of migration, with partial geographic overlap (Gómez-Bahamón et al., 2020a), suggesting that the small differences we quantified could also be driven by character displacement when migrants and nonmigrants co-occur, as opposed to correlated evolution with migratory phenotype and pressures for efficient sustained flight. Our study also revealed an evolutionary decoupling among migration and bill morphology: migration is a labile trait, while bill morphology evolved with a strong phylogenetic signal. Together, these results suggest that the effect that migratory phenotype has on the evolution of bill shape is small, compared to the effect it has on other morphological traits such as wing shape (Sheard et al., 2020).

Our phylogenetic analyses reveal that bill size and shape are tightly constrained by shared ancestry, implying close relatives resemble one another far more than distant taxa in Tyrant flycatchers. The contrast between wing and bill morphology suggests that the likelihood of correlated evolution with migration depends on how readily a trait can be remod-

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	B) PC1	**C)** Bill depth/width	**D)** Bill depth/length	**E)** Bill width/length*
Omnivores	-0.06 — 21.4	0.77 — 1.23	0.33 — 0.51	0.33 — 0.58
Invertivores	-3.80 — 15.9	0.43 — 1.23	0.29 — 0.70	0.32 — 1.25
Invertivores-herbivores	-4.65 — 12.7	0.63 — 1.14	0.35 — 0.68	0.44 — 0.80

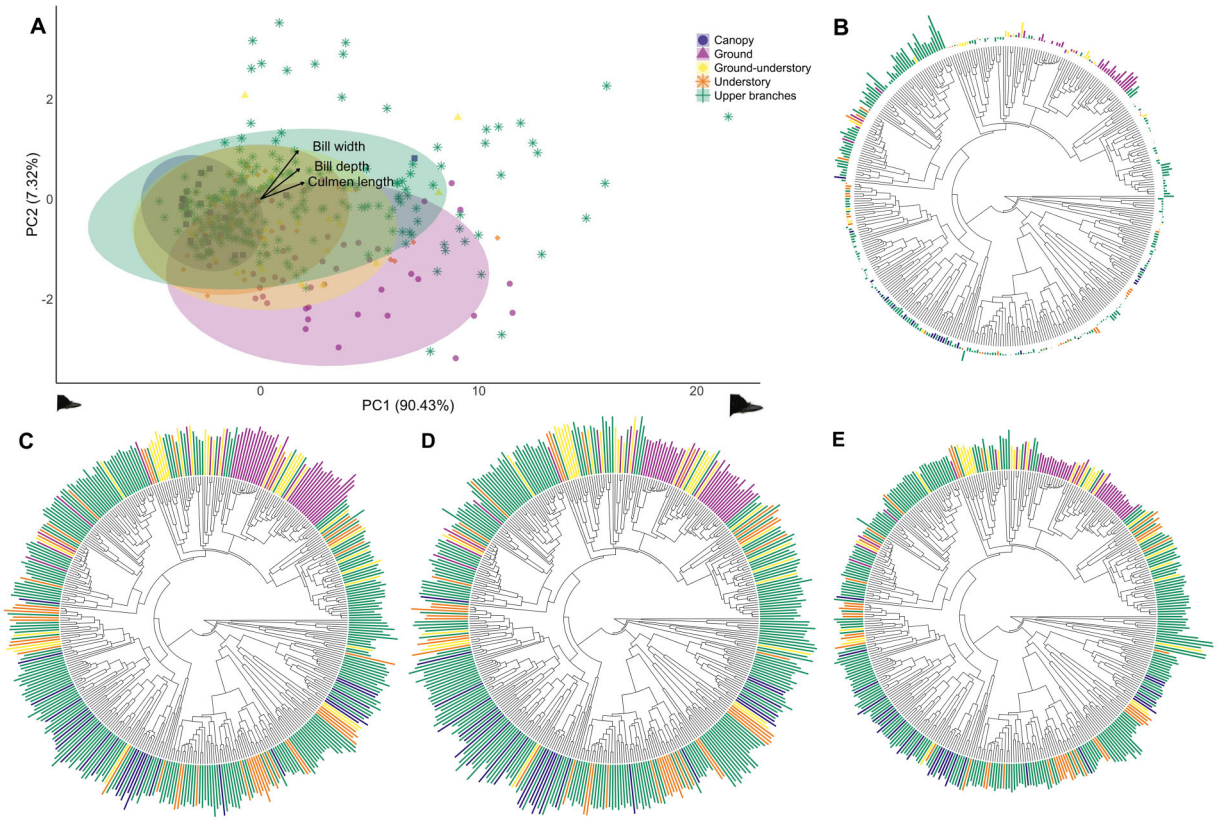
Figure 3. Tyrant flycatchers (Tyrannidae) with an invertivore–herbivore diet exhibited wider and shorter bills. (A and B) Phylogenetic Principal Component Analysis of bill sizes and (C, D, and E) tree plots of bill shapes of Tyrant flycatcher species according to diet categories (Tables 1 and S2).

Table 3. Best-fitting models (lower AIC and ΔAIC ≥ 2) analyzing score 1 from principal component analysis (PC1 = 67.3%) of bill size and measures of bill shape of Fork-tailed Flycatcher (*Tyrannus savana*) according to migratory phenotype (migratory *T. s. savana* and nonmigratory *T. s. monachus*) and tarsus length. We ran linear models and mixed linear models using age and sex as random factors, separately. We used the Yeo–Johnson transformation to normalize the dependent variables, as appropriate. Bold text indicates: ΔAIC ≥ 2 for AIC values, 95% confidence intervals (Lower–Upper CI) for CIs that do not contain zero, and $p < 0.05$ for p values. (See Table S3 for all results).

	Transf.	AIC	Lower–Upper 95% CI	*t*	*p*
PC1	No				
Null		463.4	−0.09–0.09	0.00	1.00
Tarsus + Migration | Age		**445.4**			
Tarsus | Age			**0.13–0.38**	3.90	**<.001**
Migration | Age			**−0.57 to −0.17**	−3.68	**<.001**
Depth/width	Yes				
Null		−1,640.1	0.751–0.047	1.00	.27
Tarsus + Migration		**−1,647.5**			
Tarsus			**−0.002 to −0.001**	−2.49	**.01**
Migration			**0.001–0.004**	2.75	**.01**
Depth/length	No				
Null		−904.0	**0.44–0.45**	218.5	**<.001**
Migration		−902.4	−0.006–0.012	0.66	.51
Tarsus		−903.1	−0.009–0.003	−1.07	.28
Width/length	No				
Null		−769.5	0.56–0.57	201.5	**<.001**
Migration		−769.4	−0.021–0.004	−1.38	.17
Tarsus		−768.1	−0.005–0.01	0.75	.45

eled at the molecular and developmental level. It is likely that modifying wing shape can be achieved by modest changes in the regulation of feather-growth genes, a process that could be more evolvable than the extensive remodeling required for the upper and lower mandibles that comprise the bill. Consequently, the lability of migratory behavior in Tyrant flycatchers does not seem to be limited by the selective pressures imposed by a particular feeding apparatus, diet type, or

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	B) PC1	C) Bill depth/width	D) Bill depth/length	E) Bill width/length*
Canopy	-3.63 — 7.05	0.58 — 1.14	0.37 — 0.63	0.46 — 0.75
Ground	-2.35 — 11.5	0.70 — 1.23	0.30 — 0.46	0.32 — 0.62
Ground-understory	-3.20 — 9.04	0.50 — 1.03	0.29 — 0.55	0.39 — 1.03
Understory	-3.54 — 10.9	0.58 — 1.07	0.33 — 0.56	0.34 — 0.74
Upper branches	-4.65 — 21.4	0.43 — 1.23	0.29 — 0.70	0.35 — 1.25

Figure 4. Tyrant flycatchers (Tyrannidae) that do not forage on the ground and understory showed wider and shorter bills. (A and B) Phylogenetic Principal Component Analysis of bill sizes and (C, D, and E) tree plots of bill shapes of Tyrant flycatcher species according to foraging strategy. The asterisk indicates a significant association when we ran the Phylogenetic Linear Models (Tables 1 and S2).

foraging strata. Even though migratory examples of Tyrant flycatchers shifting from insect to fruit consumption during the annual cycle exist, such seasonal diet changes may not be widespread or do not leave a detectable imprint on bill morphology, underscoring that bill evolution in this group is governed primarily by phylogenetic constraints rather than by the feeding demands of migration.

Moreover, shifts in migratory behavior might not exert a strong selective pressure that overrides the constraints imposed by other ecological needs. For instance, bill morphology has been found to be associated with correlated evolution for seed size consumption and song attributes (Demery et al., 2021; Podos, 2001), hydrodynamics in fishing species that exhibit plunge diving (Crandell et al., 2019), intra- and interspecific competitive interactions in small island passerines (Kelly et al., 2024), male–male aggressive displays in hummingbirds (Rico-Guevara & Araya-Salas, 2015), efficient preening of plumage (Clayton et al., 2005), body shapes in neotropical passerines (Brodt & Cáceres, 2025), and thermoregulation, with larger bills being more efficient at heat loss (Gamboa et al., 2024; Greenberg et al., 2012; Tattersall et al., 2009). Alternatively, the foraging demands of migrants and nonmigrants may not differ enough to drive major morphological divergence. Migrants may simply require larger quantities of similarly shaped resources rather than fundamentally different food types. In our dataset, diet explained only a weak portion of bill variation, in which species with an invertivore–herbivore diet tended to have modestly wider and shorter bills, indicating a limited effect of diet on bill shape. This weak diet-bill association is likely a consequence of the coarse categorization of diet, the multifunctionality of the bill (thermoregulation, vocalization, preening, and competition), or the strong phylogenetic signal we observed for bill traits. Future studies that track the seasonal diets of migratory species relative to their nonmigratory close relatives could clarify the true breadth and nutritional dynamics of their foraging niches.

In the present study, we asked whether migratory status explains additional variation in bill shape after accounting for coarse diet categories. Bill morphology in Tyrant flycatchers ranges from the flat, wide bills of the spade-

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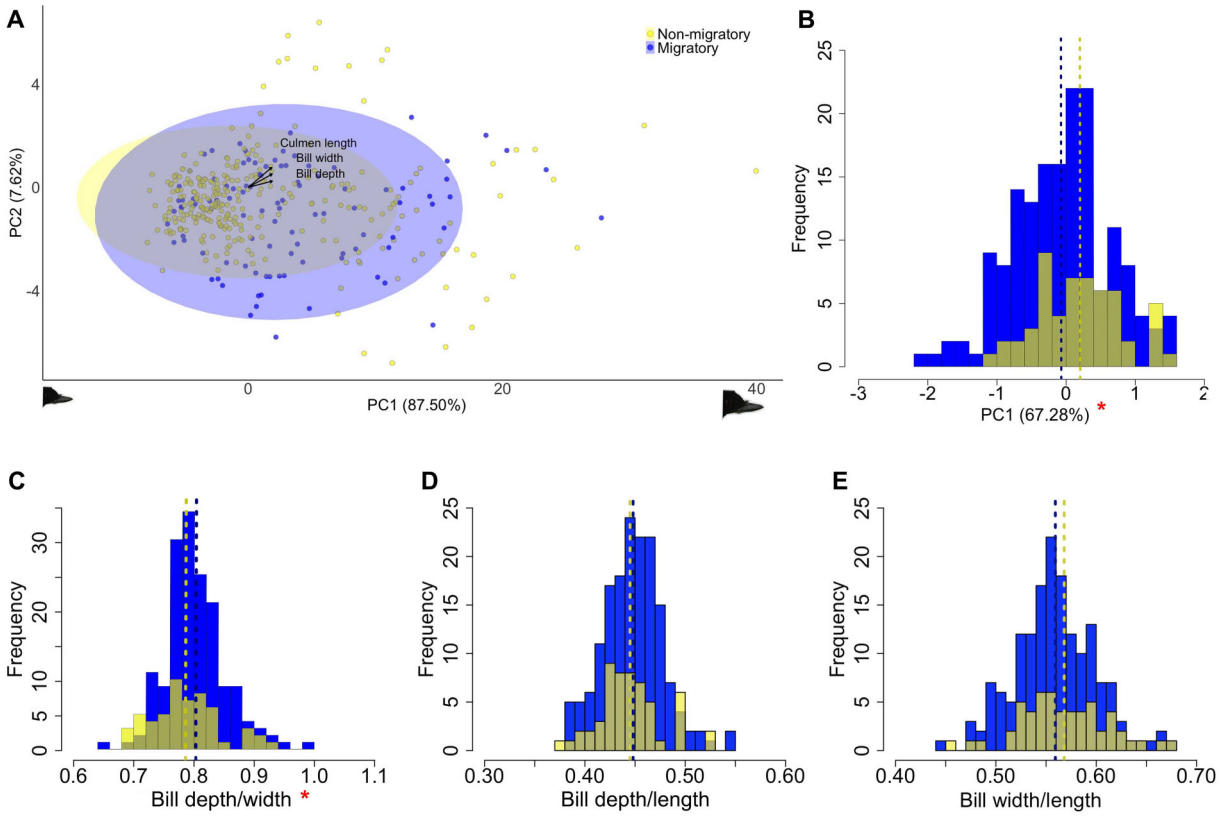


Figure 5. Fork-tailed flycatchers (*Tyrannus savana*) extensively overlapped in bill morphospace. Also, migratory Fork-tailed flycatchers (*T. s. savana*) had smaller bills and deeper and narrower bills than nonmigratory ones (*T. s. monachus*). (A and B) Principal Component Analysis of bill sizes and (C, D, and E) bar plots of bill shapes of Fork-tailed flycatchers according to migratory phenotype. Dotted lines represent the mean of measures according to migratory phenotype. Asterisks indicate significant associations when we ran linear models or mixed linear models, as appropriate (Tables 3 and S3).

bill genus (*Platyrinchus*) to the massive, slightly hooked bills of the Thick-billed kingbird (*Tyrannus crassirostris*) (Tobias et al., 2022; Winkler et al., 2020). Although most Tyrant flycatchers are insectivorous (≈ 76 %), the type and size of insects they capture can exert strong selective pressure, especially because many species live in equatorial latitudes where insect diversity and size variation are greater than in temperate regions (Chown & Gaston, 2010; Mittelbach et al., 2007; Willig et al., 2003). Consequently, the broad variation in bill morphology across Tyrant flycatchers may be driven largely by prey characteristics that our categorical diet variables only partially capture. This partial capture could underlie the pattern we observed in which insectivore–herbivore species that do not forage in the ground or understory exhibited wider and shorter bills, further emphasizing that diet–habitat interactions, more than migration, influence bill evolution.

Concordant with the interspecific level, we found extensive overlap in the bill morphospace of migrant and nonmigrant Fork-tailed flycatchers, indicating that they did not group into different specialist or generalist morphotypes. We also found that migratory Fork-tailed flycatchers had significantly smaller bills and exhibited a deeper and narrower bill shape compared to their nonmigratory counterparts (Figures 5 and S3; Tables 3 and S3). Moreover, when accounting for age, bill size (PC1) was associated with migratory phenotype, with migratory Fork-tailed flycatchers showing smaller bills. We found that bill shape and size do not differ in males and females, but factors such as age appear to contribute to intraspecific variation in bill morphology. These findings are consistent with those documented in the entire *Tyrannus* genus (MacPherson et al., 2022), and specifically in adults of *T. s. sa*vana, which exhibit differences in culmen length, regardless of sex (Rebollo et al., 2021). Previous work found no evidence of differences in diet composition between nestlings and adults in migratory Fork-tailed flycatchers (Rebollo et al., 2022), but our observations in nonmigratory Fork-tailed flycatchers suggest otherwise. Likewise, such variation could be reflecting ontogenetic changes associated with dietary size shifts, maturation, including changes in functional abilities, or developmental constraints between juvenile and adult stages (Parker, 1994; Paton & Collins, 1989; Tattersall et al., 2009). These mixed patterns suggest that bill variation within this species might reflect a combination of ecological and developmental factors beyond migration alone, such as prey size consumed during development, as well as potential niche partitioning among different age classes. Because young birds need a calcium- and protein-rich food supply to support rapid growth, and in addition juveniles must complete a rapid early-life molt, which further heightens their protein requirements, it is likely juvenile and adult Tyrant flycatchers differ in the types of insects and fruits they consume. For example, beetles and spiders tend to contain higher calcium levels than many other arthropods (Razeng & Watson 2014), and such calcium-rich items are likely over-represented in juvenile diets.

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We conclude that our findings support the hypothesis that migrants exhibited distinct bill morphology, but the effects were small and likely reflected a combination of extrinsic and intrinsic factors rather than pressures exerted by the diet breadth associated with the migratory phenotype. The extensive overlap of migrants and nonmigrants in bill morphospace at both inter- and intraspecific levels suggests that migratory phenotype was not linked with distinct specialist or generalist morphotypes as defined here. Moreover, divergence through loss or gain of migration did not result in more divergent bills, and migratory phenotype appeared to be more labile than bill morphology, suggesting they are somewhat evolutionarily decoupled.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

Data and code are available by Figshare: https://doi.org/10.6084/m9.figshare.33466462.

Author contributions

V.G.B., M.E.R., B.E.M., and C.D.C. conceived the study. V.G.B., B.E.M., A.E.J., D.T.T., and M.E.R. collected the data in the field. M.E.R. conducted the data analysis with suggestions from V.G.B. M.E.R. and V.G.B. wrote the manuscript, with substantial contributions from all other authors.

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Conflict of interest

The authors declare no conflict of interest.

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